

Bilue AI Labs

AI Engineer Technical Exercise

Candidate Brief

This exercise is your chance to show us how you think, build, and communicate about a real AI problem. We want to see judgement, not just code.

Read this pack in full before you start. If something is unclear, that is part of the test. Make a call, note your assumption, and move on.

The Scenario

You are the AI engineer on a small squad helping **FundSmart**, a fictional Australian digital consumer lender. FundSmart receives around 50 complaints a day through email, in-app messages, and call transcripts. Their six-person complaints team cannot keep up. First responses are taking two to three business days. They are breaching RG 271 service targets. Customers are frustrated. AFCA escalations are rising.

The Head of Customer Operations has asked Bilue to build an AI triage and drafting assistant. The goal is not to replace the human agents. The goal is to give every agent a solid first-pass assessment and a drafted acknowledgement the moment a complaint lands, so they can focus on judgement and resolution rather than classification and cold starts.

You have been asked to put together a working prototype and show how you would evaluate it.

Your Mission

Build a small end-to-end pipeline that takes a complaint as input and produces two things:

1. A structured triage output. Likely includes category, severity, vulnerability signals, regulatory flags, recommended routing, and SLA recommendation. The exact schema is your call, and your reasoning about the schema matters.
2. A drafted acknowledgement message that a frontline agent could lightly edit and send to the customer. Empathetic, compliant, and appropriate to the severity and vulnerability signals detected.

You must also:

- Generate your own synthetic test data beyond the seeds we provide.
- Design and run an evaluation of your system. Quantitative, qualitative, or both. Your call.
- Iterate at least once. Show us what v1 looked like, what you learned from the evaluation, and how v2 improved.
- Prepare a 30-minute presentation to walk us through your thinking and results.

What We Have Provided

Alongside this brief, you have a file called **seed_complaints.md** with three sample complaints covering different shapes and difficulties. Use these as starters. You will need to generate more test cases yourself. How you generate them, and how representative they are, is part of what we will assess.

Constraints and Permissions

Use any model, framework, or tool. AWS, OpenAI, Anthropic, open-weights, cloud-hosted, local. LangChain, LangGraph, DSPy, vanilla SDK, your own wrapper. Your call. We want to see your judgement, so be ready to explain why.

Use AI coding tools aggressively. Claude Code, Kiro, OpenCode, Cursor, Aider, Codex, whatever you use day to day. Part of your presentation is walking us through how you used AI to build this. We consider AI-native workflow a core skill for this role, not a shortcut.

Model costs are on you. If cost is a blocker, let us know and we can arrange credits.

No access to real FundSmart data. This is a synthetic scenario. Treat everything as if it were a real regulated environment.

Do not ship a production system. We are not testing UI polish, deployment, or scale. We are testing how you build a harness, generate and reason about test data, evaluate a system, and iterate.

Time Expectation

We expect this to take less than **two hours** if you are using AI tooling well. If it is taking you significantly longer, step back. Ask yourself what is slowing you down. That reflection is useful to us too, and we would rather see a tight two-hour effort with clear thinking than ten hours of polish.

If you have this brief you will be scheduled to present tomorrow. If for some reason you need to reschedule please call or email Anna Blackwell.

Definition of Done

To pass the exercise, your submission should include:

1. A harness that runs end to end complaints and produces both outputs.
2. At least 10 additional test cases you generated, with a short note on how and why.
3. At least one measured metric or evaluation result. Quantitative or qualitative.
4. Evidence of at least one iteration.
5. A 30-minute presentation.

Anything beyond this is optional. Do not over-build.

Deliverables

In the 30-minute presentation, cover:

- Framing. What did you decide the real problem was? What did you choose not to solve and why?
- Approach and harness. Architecture, schema choices, how you built it, how you used AI to build it.
- Evaluation and iteration. Your synthetic data approach, your eval method, what v1 got wrong, what v2 changed, what you would measure next.
- What you would do next, and your questions for us.

Bring questions. We learn a lot from what you are curious about.

What We Are Evaluating

Four things, roughly equal weight:

- **Harness and engineering craft.** Does your pipeline run, is the code legible, are the outputs well structured, do you have observability you can point to.
- **Test data generation.** Breadth of cases, realism, edge cases, evidence of domain empathy. Did you think about vulnerable customers, ambiguous complaints, multi-issue messages, poor English, sarcasm, emotion.
- **Evaluation design and iteration.** How do you know your system is good? What metric did you pick and why? What failure modes did you surface? What did you do about them?
- **Presentation and innovative approach.** How clearly you tell the story. How you frame the problem. And what we mean by innovative here is specific: a thoughtful, well-designed evaluation approach. Not a flashy architecture. The craft of evaluation is where Bilue sees the real differentiation.

What We Are Not Evaluating

- UI polish or a shiny front end.
- Full RAG over regulatory documents.
- Production-grade observability, security, or deployment.
- Deep FSI domain expertise. You will not be penalised for not knowing RG 271 cold. You will be penalised for not thinking about what you do not know.

A Note on Judgement

If you think this brief is scoped wrong, tell us in the presentation. If you would rather solve a different problem in this space, tell us why. We are hiring for the person who can push back constructively, not the person who nods and delivers.

Submission and Questions

Email Anna if you have any questions.

Good luck. We are looking forward to seeing what you build.